# Week 2 Problem Set: Algebraic and One-Sided Limits

Calculus Honors · Module 1: The Idea of a Limit · Lessons 1.03 and 1.04

Name: _____ Date: _____

> Show all of your work. A correct answer with no reasoning earns partial credit at most. Keep your writing neat and vertical, line up your equal signs, and label your final answers. You may check your estimates with a graphing tool, but the reasoning must be your own.

## PART A · RESOLVING 0/0 BY FACTORING

**1.** Evaluate $\lim_{x \to 4} \dfrac{x^2 - 16}{x - 4}$ by factoring and canceling. Show the shared factor explicitly before you cancel it.

**2.** Evaluate $\lim_{x \to -1} \dfrac{x^2 - 1}{x + 1}$ by factoring and canceling.

## PART B · RESOLVING 0/0 BY RATIONALIZING

**3.** Evaluate $\lim_{x \to 0} \dfrac{\sqrt{x + 9} - 3}{x}$ by multiplying by the conjugate.

**4.** Evaluate $\lim_{x \to 4} \dfrac{\sqrt{x} - 2}{x - 4}$ by multiplying by the conjugate.

## PART C · ONE-SIDED LIMITS FROM A PIECEWISE FORMULA

**5.** Let $f(x) = \begin{cases} 2x + 1 & x < 1 \\ 4 - x & x \geq 1 \end{cases}$. Find $\lim_{x \to 1^-} f(x)$ and $\lim_{x \to 1^+} f(x)$. Does $\lim_{x \to 1} f(x)$ exist? If so, state its value.

**6.** Let $g(x) = \begin{cases} x + 3 & x < 2 \\ x^2 & x \geq 2 \end{cases}$. Find $\lim_{x \to 2^-} g(x)$ and $\lim_{x \to 2^+} g(x)$. Does $\lim_{x \to 2} g(x)$ exist? Explain.

## PART D · READING EXISTENCE FROM A GRAPH

**7.** The graph of $y = h(x)$ approaches height 2 from both the left and the right of $x = 0$, where an open circle is drawn, and a separate solid dot marks $h(0) = 5$. State $\lim_{x \to 0} h(x)$ and $h(0)$, and explain why they are allowed to differ.

**8.** A step graph holds steady at height $-1$ for $x < 3$, then jumps to height 4 for $x \geq 3$, with a solid dot at $(3, 4)$. State $\lim_{x \to 3^-}$, $\lim_{x \to 3^+}$, and $\lim_{x \to 3}$, and explain your reasoning for the last one.

## PART E · MIXED REVIEW AND EXPLANATION

**9.** Evaluate $\lim_{x \to 2} \dfrac{x^3 - 8}{x - 2}$. Name the technique you used and why it applied here.

**10.** Convince a classmate: before you factor and cancel a fraction, why must you first check that the numerator and denominator both approach zero? What could go wrong if you cancelled without checking, on a fraction whose denominator's limit is not zero?